WHAT THE OTHERS SAY

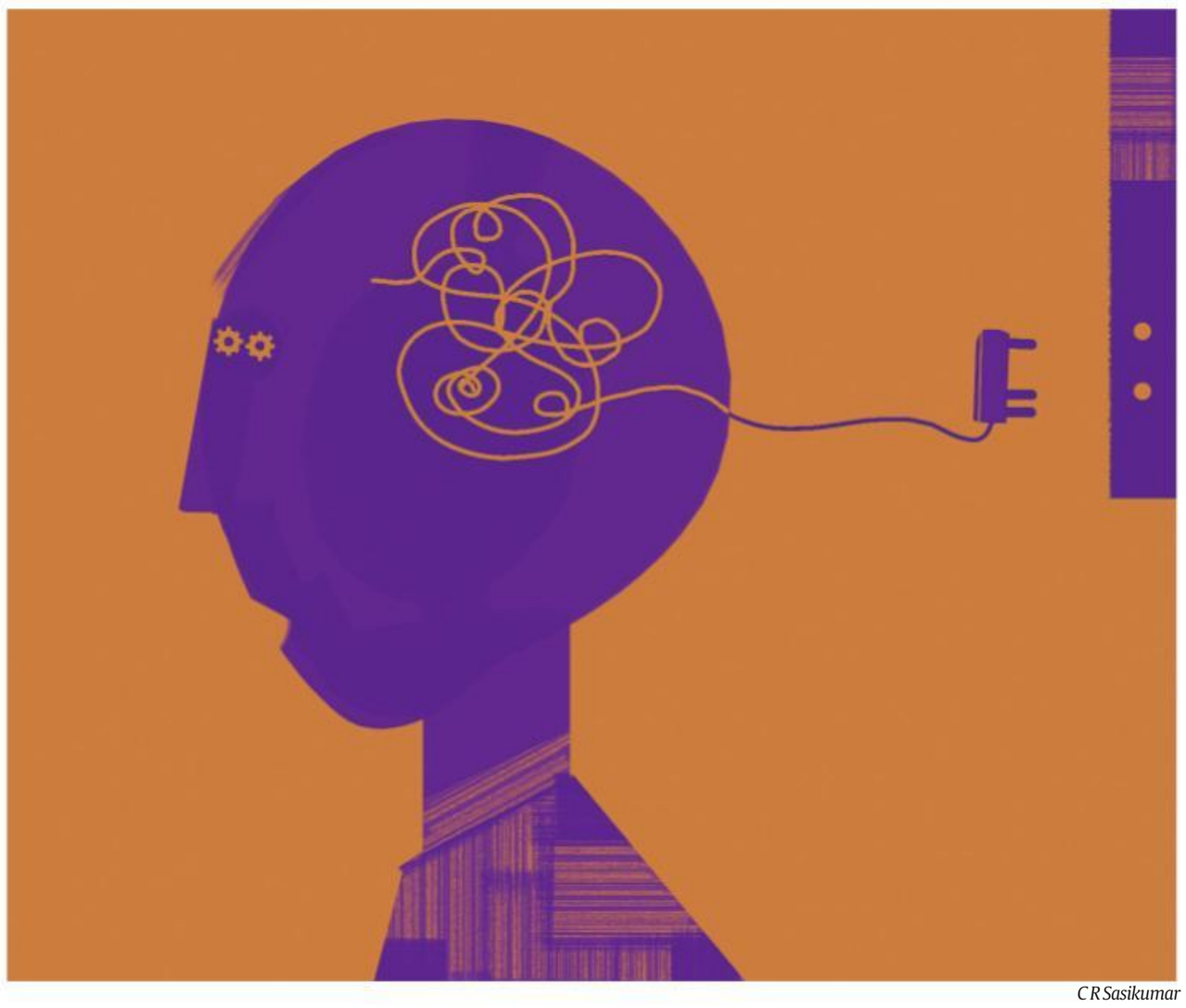
"Unfortunately, successive governments (in Pakistan) have viewed even basic human rights as acts of charity to be dispensed at will. This mindset is reflected in practically all spheres of life, including the justice system itself." —DAWN

A syllabus for doers

India needs an engineering education that is not detached from people's lived reality — but one that can throw up solutions that provide jobs, save resources and improve lives



PRIYA JADHAV



C R Sasikumar

A RECENT ARTICLE by economist Rathin Roy points out that action on climate change globally has become about green technologies, rather than lower consumption. It says that keeping the focus on reducing inequalities among people will naturally result in more environmental policies. An analogous situation prevails in the area of education in India.

The instruction imparted in schools today is beyond the grasp of most students, and is detached from their lives. Students in Class 10 are expected to sum up series and parallel resistances, and conceptually grasp the workings of a DC motor. The reality is that less than 50 per cent of Class 8 students of rural government schools can do division (ASER 2016–18).

This is true of engineering education too and affects decision-making capacity in the country. It happens in two ways. Directly, a difficult engineering education that is detached from people's lived reality, leads to poor intellectual development. It also ends up leaving unattended numerous gaps that could lead to development and the creation of jobs. Let us look at some examples in the area of electricity supply for irrigation.

The woes of irrigation energy are well documented. A simple web search will show subsidised electricity, inefficiency, groundwater depletion and poor power quality as commonly used terms. Recently, there has also been discussion about direct benefits transfer instead of subsidised electricity, and the use of solar PV pumps as a solution.

An accepted narrative is that subsidies disincentivise distribution companies from investments to improve the quality of supply. However, on investigating, we find that there are, in fact, many solutions that can reduce the cost of supply, improve quality, and also generate employment. This was shown through an investigation conducted by CTARA at IIT Bombay, under the Project on Climate Resilient Agriculture, Government of Maharashtra.

An exercise was done to document the distribution network in Umbarda Bazaar, a typical village in Washim district. The village has 190 pump connections. A simple set of rules was applied to find a restructured design of the network, which had grown somewhat haphazardly over the years, to provide connections to 59 unconnected wells, and also improve performance. The implementation could save Rs 12-15 lakh in new connection costs, and Rs 2.5 lakh per year in repair costs of pumps and transformers. It would cut down distribution losses, and reduce the stress and agricultural loss that farmers suffer due to breakdowns. The case is representative of villages in Maharashtra.

Such redesigning can be done by an electrical engineer with basic capabilities. The lines were to be moved as a part of a restructuring process, one that could provide employment to labourers. This shifts the focus from new materials and infrastructure to human capabilities.

A more detailed GIS-based system incorporating cropping, and electrical systems could be used to optimise such decision-making further. While this needs more sophisticated systems, it does not need cutting-edge research capability in any one discipline. Instead, it warrants a new approach that integrates physical systems across disciplinary boundaries.

Another solution involves groups of 25-30 farmers following a schedule, such that pumping times are staggered even while satisfying irrigation requirements. Such a practice improves the quality of supply. The schedule could be made as a Sudoku-like problem of filling a table based on some thumb rules, or through an optimisation problem clubbed with power systems analysis. Once more, maximising infrastructure use through human capability.

This example illustrates a move away from designing for perfect outcomes for users, to a less than perfect one where people have to adjust to the system. Or, at least the system design could account for locally expected usage. However, such adjustments ought to be formalised — this is not *jugaad* where farmers themselves decide that half the group will operate on alternate days. That is a suboptimal use of the system where people are abandoned to do the best they can when instead our engineers could help them create a better solution.

It may be noted that these examples are very specific to India, and require different skill levels, all the way from diploma holders to PhDs. Hence, solutions that provide jobs, save resources, and improve the lives of our people, need us to investigate issues at the ground level and account for local conditions. Broad-brush targets such as 18 lakh solar pumps (PM-KUSUM) only lead to the organisationally most convenient implementation, not the best outcomes for a certain capital.

The instruction imparted in schools today is beyond the grasp of most students, and is detached from their lives. Students in Class 10 are expected to sum up series and parallel resistances, and conceptually grasp the workings of a DC motor. The reality is that less than 50 per cent of Class 8 students of rural government schools can do division (ASER 2016-18). This is true of engineering education too and affects decision-making capacity in the country.

So, how is this untapped potential connected to better education? It is generally accepted that familiar contexts lead to better learning. Nilesh Nimkar, an educationist working in tribal areas, notes that if forced to choose, a local context may be more important than tribal language as a medium.

An MIT report, "The global state of the art in engineering education", has studied some institutes that are adopting a new pedagogy for emerging problems. The recurring themes are: Contextualised learning, application to real engineering problems, hands-on learning, regional problems, and inter-disciplinarity.

All of this indicates a move away from deep theoretical education to a practical broad-based one. Currently, students may do a final-year project on speed control of a motor, but graduate without understanding the benefits of power factor control in motors. Projects like designing bus routes for optimal service or comparing solar PV pumps to the grid for a crop system, do not result in new technology but contribute to its deployment. And result in the type of engineering training that is sorely needed.

There is a trend to address every problem from climate change to healthcare with the likes of artificial intelligence and blockchain. As a result, we are neglecting the education of an overwhelming majority, and missing out on addressing a large problem space.

The writer is assistant professor, Centre for Technology Alternatives for Rural Areas, IIT Bombay

Mission for rejuvenation

By restoring ancient centres of Hinduism, PM Modi gives shape to an agenda of cultural and spiritual transformation



SHRI M

INDIA IS A land of knowledge, art, culture, science and spirituality. For centuries, India has shown the world the path of humanism and global peace. The idea of "Vasudhaiva Kutumbakam" (the world is one family) is at its core.

Since the Vedic period, our holy land had faced various onslaughts. But our rich art and culture, our religion and the ancient treasure troves of wisdom, philosophy, science and literature have survived under the watch of our great saints and sages over the centuries.

Since my early years, I have been seeking the "truth", the "eternal" and the power of spirituality. In my quest for truth and enlightenment, I choose the Himalayas as my abode. I visited the holy Kedarnath Dham on many occasions, and it was here that I heard the amazing story of how the great Maratha queen Devi Ahilyabai Holkar reconstructed important Hindu shrines and temples at several places, including at Gangotri, Rameswaram, Dwarka and Gaya. Some of the famous temples which were revived under her patronage include Kedarnath Dham, Shri Omkareshwar, Kashi Vishwanath, Jagannath Puri and Somnath.

My spiritual quest took me to many places across India. I went to several temples which were renovated by the great Maratha queen. My earnest desire was to dive deep into the world of spiritualism, my quest was to know myself and learn the country's ancient wisdom. That took me to the doors of Shri Kashi Vishwanath where I was initiated into new dimensions of spiritual energy.

During my stay in Kashi, I pondered over who will carry forward Ahilyabai's legacy of rejuvenating and reconstructing our great centres of religion and spiritualism, the ancient temples which were plundered by foreign invaders. The thought that our religious and spiritual legacy, our ancient art and culture, and the great centres of Hinduism were gradually degenerating perturbed me.

Now, as I watch the renovated and rejuvenated Baba Vishwanath Dham in the eternal city, Kashi, being dedicated to the world, my inner self is at peace as I know that some true sons of the soil are still there to look after the preservation, conservation and promotion of Sanatan Dharma, to safeguard and preserve our ancient centres of spiritual energy. Prime Minister Narendra Modi is one such leader. Under his leadership, India is moving ahead in meeting its physical as well as spiritual goals as we celebrate our Azadi Ka Amrit Mahotsav. I am thrilled at the very thought of getting a glimpse of Maa Ganga from Shri Kashi Vishwanath Dham.

Despite facing several challenges, Ahilyabai Holkar left no stone unturned to safeguard, rebuild and lay a strong foundation of Indian culture, spiritualism and Hinduism. Narendra Modi is walking in her footsteps by taking up the transformation and rejuvenation of our great centres of spiritualism and religion. I say this as I witness the rebuilding and transformation of Kedarnath Dham and the reconstruction of the *samadhi sthal* of Adi Shankaracharya, the great seer and teacher.

No one ever thought that the complex issue of the Ram Temple in Ayodhya will be resolved in such an amicable and smooth manner. Today, work on the construction of a grand temple is underway. To whom does this credit go? Narendra Modi is also instrumental in setting up several centres of spiritualism and religion abroad.

One important initiative of Prime Minister Narendra Modi worth mentioning relates to the global recognition of our ancient science of yoga. Yoga, which is an integral part of our culture and our spiritual lifestyle, is today healing and helping crores of people, not only in India but also overseas. It has become a way of life for people across the globe. For those who do not follow the path of spiritualism, it has become a tool to improve their physical and mental health. Even modern science has acknowledged the benefits of yoga.

I started doing yoga and *pranayam* when I was 10, and it is because of this practice that today I am in good physical and mental health. People have realised and felt the real power of yoga and *pranayam* during the pandemic which helped in healing and keeping them mentally and physically fit during a challenging time. India and the world will always remain indebted to PM Modi for his concerted efforts in reintroducing the science of yoga to the world by persuading the UN to proclaim International Yoga Day.

India is going through a cultural and spiritual transformation. In his monthly radio talk, "Mann Ki Baat", PM Modi referred to Brazil's Jonas Masetti and highlighted his work and accomplishments, especially the lessons on Vedanta and the Bhagavad Gita that he has been giving near Rio de Janeiro. The PM also called upon non-resident Indians to play their part in global peace and in promoting Indian ethos and values.

The social and cultural impact that PM Modi has made on the lives of crores of Indians through his various initiatives, despite facing severe challenges and hardships, will be etched in our history forever. These include getting the poor access to proper toilets, Swachh Bharat Abhiyan, Ujjwala Yojana, Beti Bachao Beti Padhao, water conservation, Namami Gange, and so on. The great success of these programmes is nothing short of the social, cultural and spiritual rejuvenation of the citizens of India.

The writer is a spiritual guru and thinker

Network of terror

Political churn in Af-Pak region opens up space for extremist outfits



ANJU GUPTA

THE TAKEOVER OF Afghanistan has created a huge challenge for the Taliban as much as for the world. Since August 15, the Islamic State Khorasan Province (ISKP) attacks on Taliban and Afghan targets have increased sharply and so have the Tehrik-e-Taliban Pakistan (TTP) attacks on Pakistani forces across the border. While the Taliban denies that ISKP has a "serious" presence in Afghanistan, Pakistan claims that the TTP is operating from Afghanistan. The two groups have claimed hundreds of attacks, including complex coordinated suicide and targeted attacks — all pointing to the existence of robust terrorist infrastructure in the Af-Pak region. In addition, hundreds of targeted killings of ex-Afghan National Defence and Security Forces (ANDSF) officers and local leaders are being blamed on the Taliban, opening the possibility of new local militias forming against the Taliban and adding to the insecurity.

Several irritants have surfaced in Taliban-Pakistan relations, with the latter arbitrarily closing the border crossings; meting out bad treatment to Afghans at the borders; continuing the fencing of the border to separate Pashtuns; denying diplomatic recognition to Taliban, and failing to get financial support from China for Taliban. However, the most embarrassing "truth" for the Taliban is the "invisibility" of Amir-ul-Momineen (AuM), Haibatullah Akhundzada, to thousands of cadres who, having pledged allegiance to him, have fought to establish "true" Islamic rule under him. Media reports suggest that he was last seen in a Quetta mosque and may actually be a "house guest" in Pakistan.

Soon after the takeover, the TTP repledged allegiance to the AuM. Pakistan has failed to drive a wedge between the two groups. Since the Doha Agreement of February 2020, the TTP network upped the ante and merged at least 12 major groups in August, under Noor Wali Mehsud. Emulating the Taliban, the TTP has been positioning itself as a nationalist group against the "un-Islamic" Pakistan army and has demonstrated its prowess by attacking Pakistani forces at will, mostly across the tribal belt and in cities like Quetta and Islamabad.

In 2020, the TTP claimed more than 180 attacks and over 400 casualties to Pakistani forces. This year, up to October, the number of attacks claimed has gone up to about 240. Between August and October, TTP has claimed over 120 attacks and 400 casualties to Pakistani forces. Additionally, the Baloch groups have claimed over 50 attacks and 100 casualties to the Pakistani forces during July-September. In 2021, at least three big attacks on Chinese interests in Pakistan have further jeopardised Pakistan's efforts in bringing China "deeper" into Pakistan and Afghanistan.

Facing the onslaught of the TTP, the Pakistan army first engineered its proxy Pashtun groups such as the Hafiz Gul Bahadur group to declare a ceasefire to "show progress", while it continued drone strikes and targeted killings in the tribal belt. With no foreign intelligence agencies to blame and the Taliban in power in Afghanistan, TTP attacks are an embarrassment for the Pakistan army, which has even stopped admitting its casualties, except for senior ranks. The Pashtun Tahafuz Movement has managed to successfully tap the anti-army sentiment in the tribal belt, making it very difficult to start large-scale operations either. The perilous state of Pakistan's economy and grey listing by FATF are additional deterrents.

The Pakistan army also unleashed proxy violence in Afghanistan through the ISKP, including complex coordinated attacks in big cities, and attacks on Taliban leaders. It is widely believed that since the early days, the command and control of the ISKP have been under the control of the Pakistan Army, while fighters continued to be recruited in the name of "Daesh in Khorasan". More than 400 ISKP cadres locked in Afghan prisons were released when the Taliban took over. The ISKP also continues to claim targeted killings in Pakistan's tribal belt.

In early October, Pakistan announced that the Taliban was mediating talks with the TTP in Afghanistan and has, of late, offered many concessions, including permission to the Taliban to occupy Afghan missions in Pakistan, presenting Taliban's foreign minister as a state guest and releasing regular drips of humanitarian assistance. On November 9, Pakistan and the TTP declared a month-long ceasefire. But the very next day, the Taliban rejected Pakistan's assertion that it would militarily act against the TTP, should the peace talks fail.

The month-long ceasefire of TTP ended on December 9 and since then, it has claimed five attacks on Pakistani forces. Meanwhile, Taliban spokesperson has called Pakistan system "un-Islamic" and in a video, TTP has now described itself as a branch of Taliban. Further escalation in violence would abet recruitment by two groups, including from thousands of Taliban fighters, criminals and al Qaeda cadres released from Afghan prisons. The TTP would recruit cadres from Pakistan's tribal belt and Afghanistan, while the ISKP would target cadres from South Asia and beyond, using some to carry out lone wolf attacks in their own countries.

The Taliban possesses raw muscle, but does not have professional counter-terrorism skills or the tools needed to understand or address violence in Afghanistan. Its machinery is killing those suspected to be "enemies" or "spies" and claims are made in the name of criminals or ISKP cadres. As the Taliban further fails in providing security, and the humanitarian crisis deepens, individual commanders may start colluding with criminals and smugglers of drugs, humans, goods and weapons to generate funds. The ungoverned pockets of Afghanistan, escalating violence, and mercenary commanders would present an opportunity for international groups like al Qaeda and Daesh to relocate and strengthen their regional and global operations from Afghanistan.

The writer is an IPS officer. Views are personal

LETTERS TO THE EDITOR

MATTERS OF BIAS

THIS REFERS TO the article, 'Just expectation' (IE, Dec 13). The very act of adjudication in a court of law between the contesting parties could make the losing party unhappy and critical of the decision, even if it is completely objective and based exclusively on the merits of the case. So, the judiciary ought to take such criticism as an occupational hazard. However, Shashi Tharoor's criticism of higher judiciary for allegedly failing to protect the constitutional rights of the citizens is questioning the lack of pro-active action of adjudication against the laws that are perceived by sections of society through their petition to be ultra vires of the Constitution. The apparent delay in taking up these cases like the abrogation of Article 370, the Citizenship Amendment Act, and electoral bonds, which all have multidimensional ramifications for the health of the nation's democracy, seem to let down citizens on their just expectation. There are interesting findings in recent studies on bias among judges in lower judiciary, that have drawn opposite conclusions necessitating the need for further research.

L R Murmu, *Delhi*

BONDS OF LOVE

THIS REFERS to the report, 'First tranche of medical aid from India after Taliban takeover reaches Kabul'(IE, December 12). The NDA government has given proof of its benevolence by supplying a 1.6-tonne consignment of emergency life-saving medicines for a children's hospital. This supply indicates more generosity as the medicines were not solicited by the Taliban government. But we must not expect reciprocity from this war-ravaged country. It will be enough for Indians if Afghanistan's children start considering every Indian supplier of medicines for them as Dilliwala out of love

Tarsem Singh, *Mahilpur*

AFTER PROTESTS

THIS REFERS TO the editorial, 'Beyond barricades' (IE, December 10). A concerted effort and sacrifice made by the farmers have brought a fruitful outcome and as the SKM claims, most of their demands have been met. But the issue of farm reforms and enhancing farmers' income remains a pertinent question and the government needs to ensure that it isn't put on a backburner. The committee to decide upon the demands of a legal MSP must constitute varied stakeholders including subject experts, and not just the politicians. States also need to pitch in with suggestions for fundamental transformation of the farm sector like dependence on monsoon, stubble burning and shouldn't bank on farmers' votes via poll sops.

Nischai Vats, *Sultanpur*

SHORT ON

THIS REFERS TO the article, 'Time For the Future' (IE, December 12). The author asserts that Bangladesh has maintained secularism as a state principle. However, he fails to mention that there is a curious contradiction in the Bangladesh constitution (article 2A) in declaring Islam as a state religion. This is indicative of the unhealthy relationship between state and religion in modern Bangladesh, which has defied many odds to flourish and prosper.

Anjan Ray, *Delhi*